

EC2 : Adding Volumes & Creating a Text File

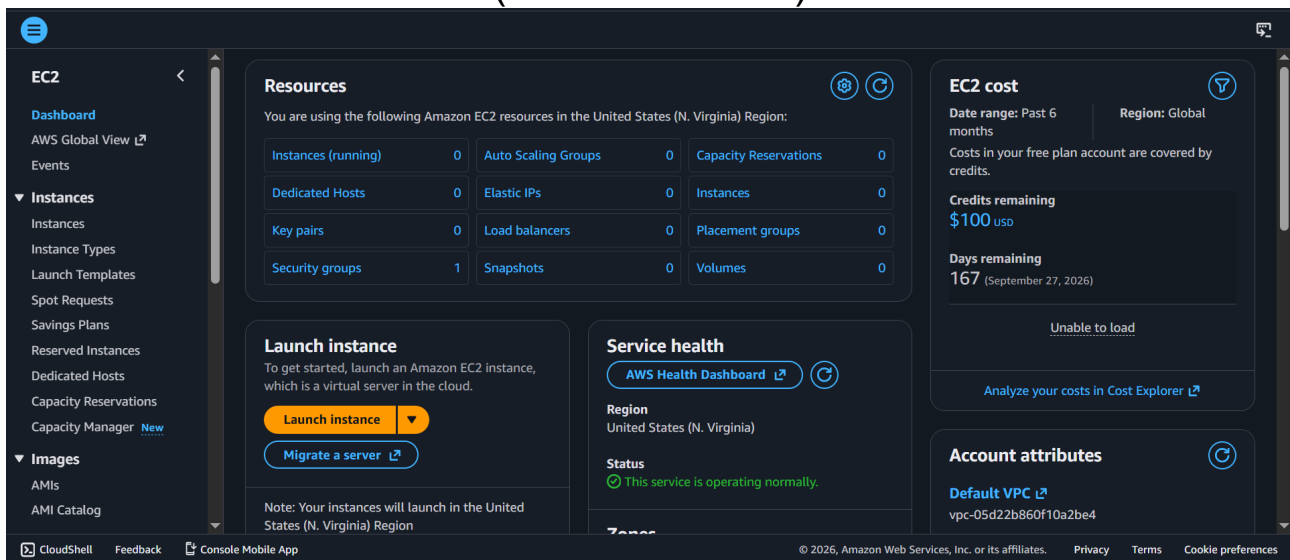
Amazon EC2 (Elastic Compute Cloud) is a service provided by Amazon Web Services that allows users to run virtual machines (instances) in the cloud. These instances can be extended with additional storage using volumes.

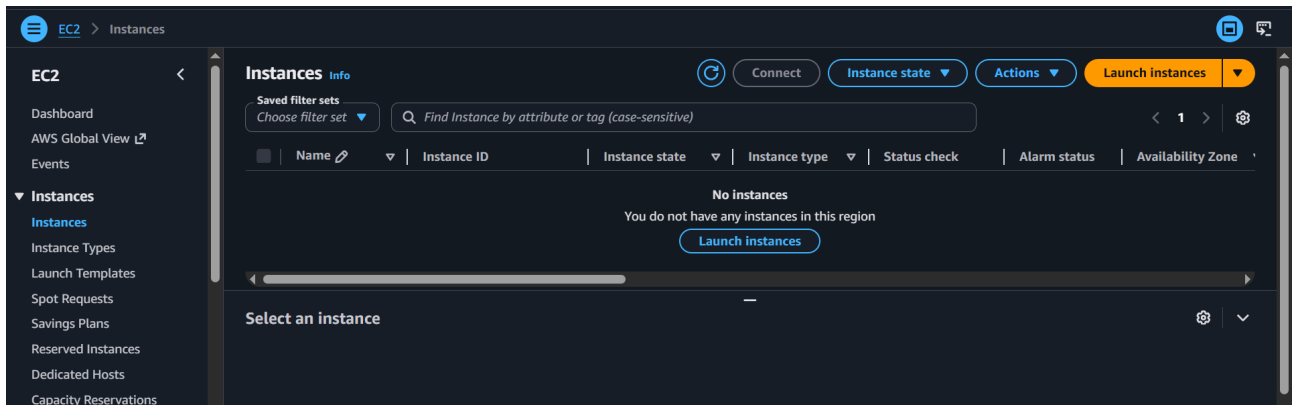
Objectives:-

- Add a new storage volume to an EC2 instance
- Attach and mount the volume
- Create and store a text file in the volume

Step 1 : Launching an Instance

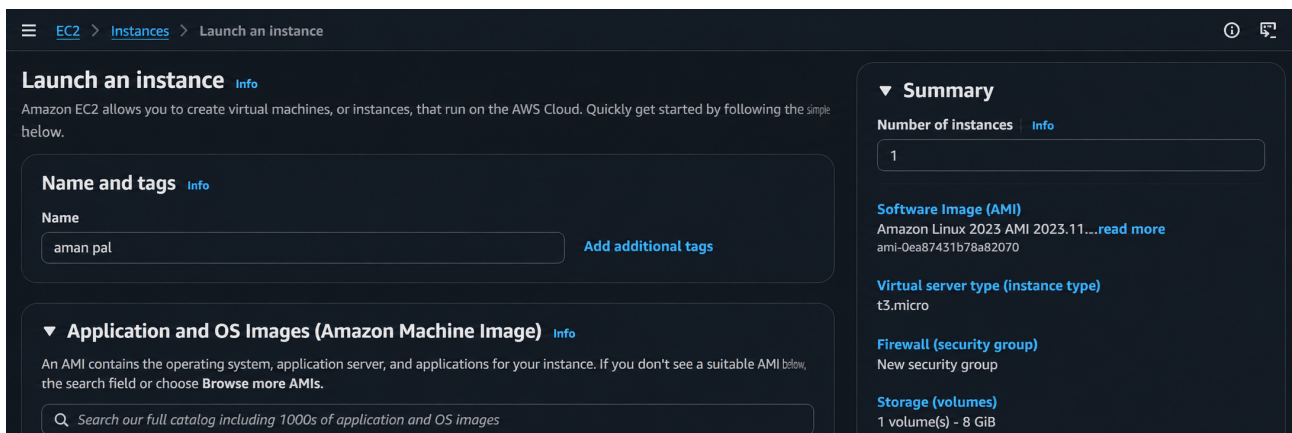
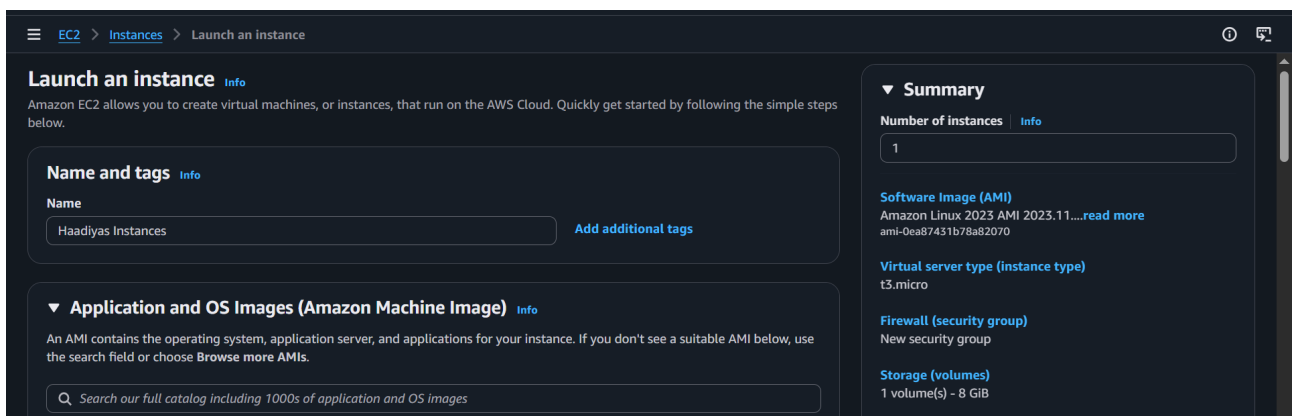
Log into AWS Management Console. Navigate to Amazon EC2. Click on **Launch Instance**. This starts the process of creating a virtual server in the cloud (a.k.a : instance).





Step 2 : Launching an Instance

Here you can provide a name for your instance(e.g., *Haadiyas Instances*) and choose an Amazon Machine Image (AMI), which defines the operating system (e.g., Amazon Linux) and pre-installed software for the virtual server.



Step 3 : Select Instance Type

Now, choose the instance type, which defines the CPU, memory, and performance capacity of your virtual server.

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-0ea87431b78a82070 (64-bit (x86), uefi-preferred) / ami-088cedaa951dccc6a5 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.11.20260406.2 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username
64-bit (x86)	uefi-preferred	ami-0ea87431b78a82070	2026-04-06	ec2-user

Verified provider

▼ Instance type Info Get advice

Instance type

t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0104 USD per Hour On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Summary

Number of instances Info

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.11.....read more
ami-0ea87431b78a82070

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Preview code

Step 4 : Configure Key Pair and Network Settings

Select or create a key pair for secure login and configure network settings like VPC, subnet, public IP, and security group rules (SSH, HTTP, HTTPS) to control access to the instance.

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Proceed without a key pair (Not recommended) Default value Create new key pair

▼

Network settings

Info

Edit

Network

Info

vpc-05d22b860f10a2be4

Subnet

Info

No preference (Default subnet in any availability zone)

Auto-assign public IP

Info

Enable

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

☒ Allow SSH traffic from

Helps you connect to your instance

Anywhere
0.0.0.0/0

☒ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

⚠

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

×

Step 5 : Configure Storage

Set the storage size (e.g., 8 GiB) and volume type (gp3) for your instance, which determines how much data your server can store.

▼

Configure storage

Info

Advanced

1x

8

GiB

gp3

▼

Root volume, 3000 IOPS, Not encrypted

Add new volume

🔄

Click refresh to view backup information

↻

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

File systems

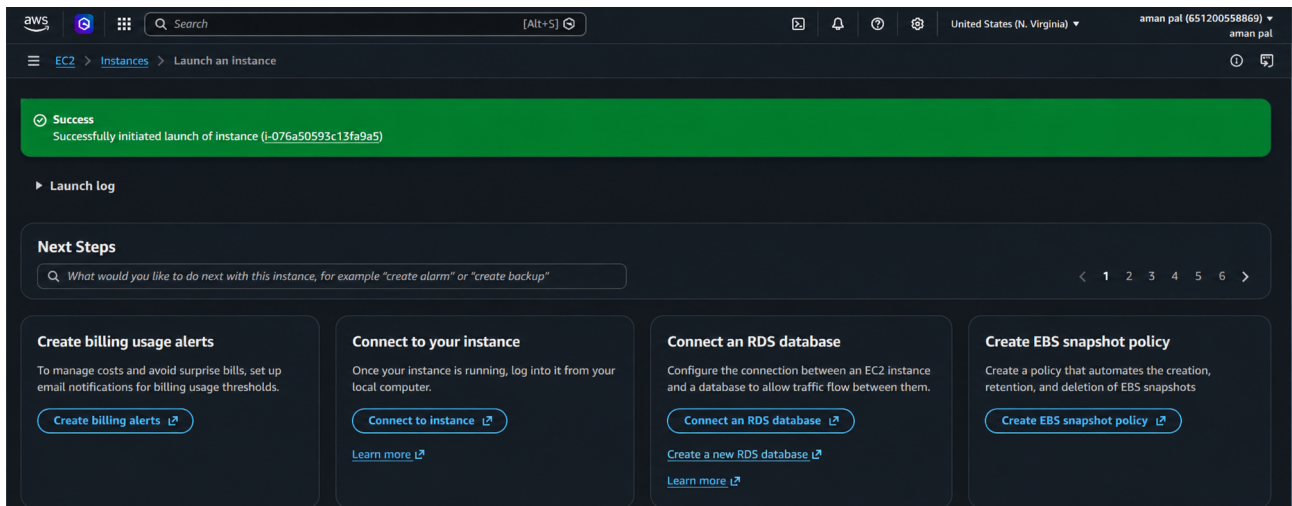
☐ S3 Files - new

☐ EFS

☐ FSx

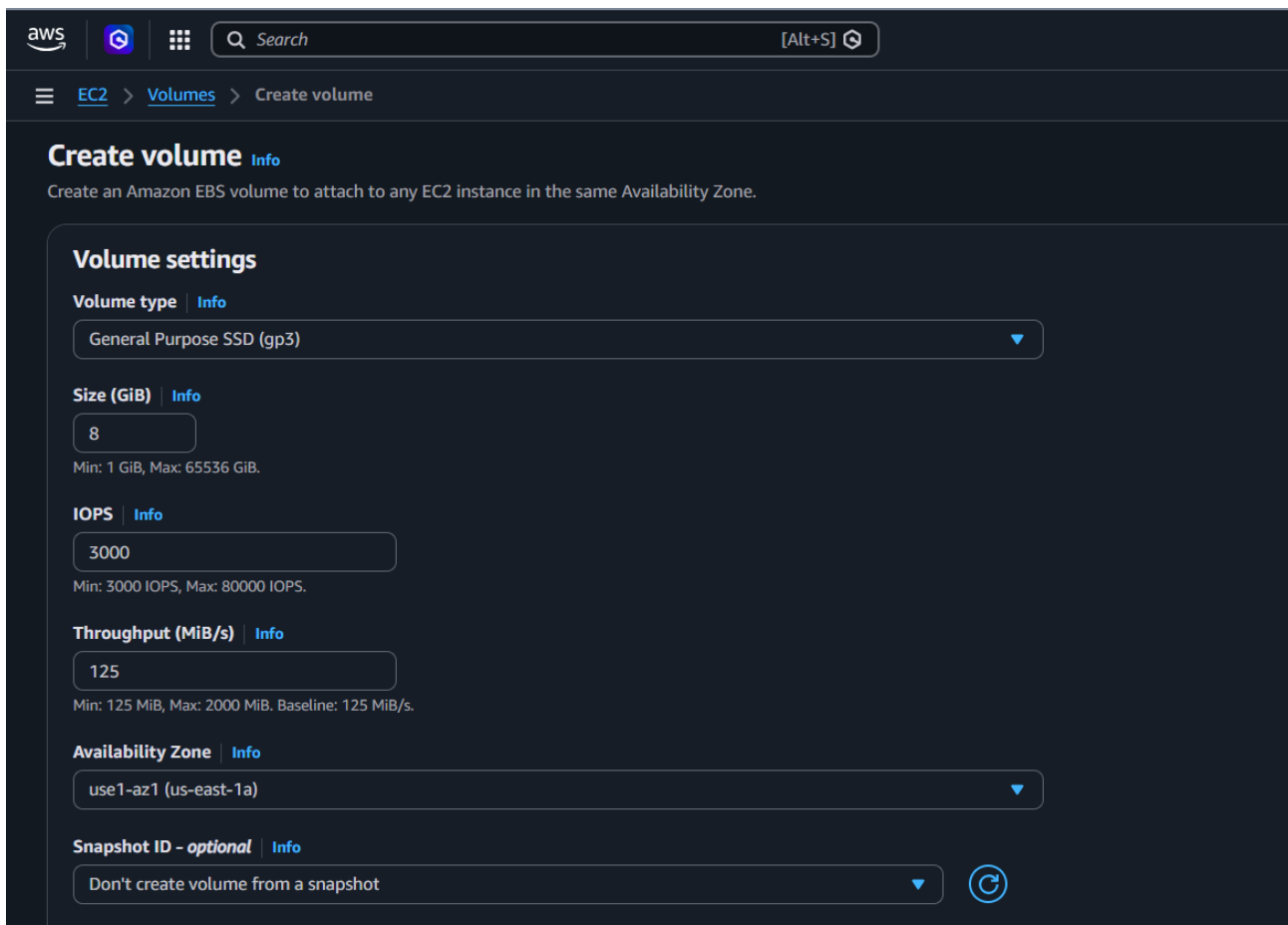
Step 6 : Storage Instance Successfully Created

After clicking “Launch Instance,” the system shows a success message indicating that your EC2 instance has been created and is now being initialized in the cloud.



Step 7 : Create an EBS Volume

Set up the EBS volume by selecting **volume type** (gp3), **size** (e.g., 8 GiB), **IOPS**, throughput, and **availability zone**, which defines the storage performance and location for your instance.



Step 8 : Viewing the Details

Here, select the running instances to **view** its details and other **configuration** information.

The screenshot shows the AWS Management Console interface for a specific volume. At the top, a green notification bar states "Successfully created volume vol-07a1be069feb1b077.". Below this, the "Volumes (1/1)" section displays a table with one volume: vol-0216746befe21b667, type gp3, size 8 GiB, IOPS 3000, and throughput 125. The "Details" tab is selected, showing various attributes: Volume ID (vol-0216746befe21b667), Size (8 GiB), Type (gp3), Status check (Insufficient data), Volume state (In-use), IOPS (3000), Throughput (125), Fast snapshot restored (No), Availability Zone (use1-az4), Created (Mon Apr 13 2026 12:54:50 GMT+0530), Multi-Attach enabled (No), Attached resources, Outposts ARN, Managed, and Operator.

Volume ID	Size	Type	Status check
vol-0216746befe21b667	8 GiB	gp3	Insufficient data

Volume state	IOPS	Throughput
In-use	3000	125

Fast snapshot restored	Availability Zone	Created	Multi-Attach enabled
No	use1-az4 (us-east-1c)	Mon Apr 13 2026 12:54:50 GMT+0530 (India Standard Time)	No

Attached resources	Outposts ARN	Managed	Operator

Step 9 : Connect the Instance

Choose the connection method and click **“Connect”** to access the virtual server directly from the browser itself.

The screenshot shows the "Connect to instance" page in the AWS Management Console. The instance ID is i-076a50593c13fa9a5 (aman pal). The VPC ID is vpc-05d22b860f10a2be4. The security groups are sg-046cb0d4719aba44e (launch-wizard-1). The IAM role is -. The "EC2 Instance Connect" tab is selected, showing the "Connection type" section with two options: "Connect using a Public IP" (selected) and "Connect using a Private IP". Under "Public IP", the "Public IPv4 address" is selected with the value 13.220.82.7. The "Username" section is also visible, with the default username "ec2-user" entered. A note states: "Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username." The "Connect" button is highlighted in orange.

Instance ID: i-076a50593c13fa9a5 (aman pal)

VPC ID: vpc-05d22b860f10a2be4

Security groups: sg-046cb0d4719aba44e (launch-wizard-1)

IAM role: -

Connection type:

- ☒ Connect using a Public IP
Connect using a public IPv4 or IPv6 address
- ☐ Connect using a Private IP
Connect using a private IP address and a VPC endpoint

Public IPv4 address: 13.220.82.7

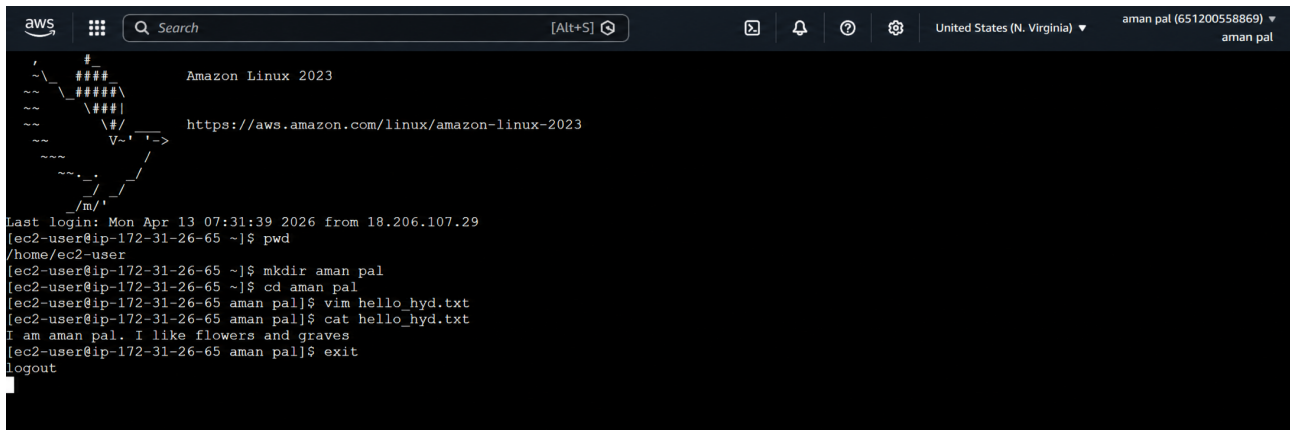
Username: ec2-user

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel Connect

Step 10 : Execute Commands in Instance Terminal

Use the terminal to run Linux commands like creating directories (mkdir), navigating folders (cd), creating files (vim), and displaying content (cat) to interact with the EC2 instance.



```
aws
Search [Alt+S]
United States (N. Virginia) aman pal (651200558869) aman pal

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Mon Apr 13 07:31:39 2026 from 18.206.107.29
[ec2-user@ip-172-31-26-65 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-26-65 ~]$ mkdir aman pal
[ec2-user@ip-172-31-26-65 ~]$ cd aman pal
[ec2-user@ip-172-31-26-65 aman pal]$ vim hello_hyd.txt
[ec2-user@ip-172-31-26-65 aman pal]$ cat hello_hyd.txt
I am aman pal. I like flowers and graves
[ec2-user@ip-172-31-26-65 aman pal]$ exit
logout
```